# Hierarchical Cross-Modal Feature Shuffling Network for Unsupervised Multimodal Remote Sensing Clustering

Shufang Xu, *Member, IEEE*, Ruizhe Liu, Xinchen Ding, and Hongmin Gao, *Senior Member, IEEE*

***Abstract*—Current fusion methods often treat multimodal features equally or rely on simple concatenation, failing to capture the complex interdependencies between modalities. Furthermore, most approaches process features at a single scale, neglecting the hierarchical nature of visual information in remote sensing (RS) data. To address these issues, we propose a hierarchical cross-modal feature shuffling network (HCFSN). First, the local patch shuffling (LPS) module introduces a weak local perturbation with a shared shuffling order across modalities, reducing overreliance on fixed absolute positions while preserving paired spectral–geometric content. Second, the dynamic group shuffling (DGS) module facilitates adaptive channel-wise interaction through a data-driven gating network, which dynamically computes interaction weights between semantic channel groups to circumvent manual rule constraints. Third, the global cross-modality attention (GCMA) module enables fine-grained global cross-modal interaction by: 1) establishing long-range spatial dependencies via global attention mechanisms; 2) preserving spatial alignment through spatial–channel dual attention; and 3) supporting cross-modal information interaction with multihead attention to capture diverse semantic associations. Fourth, the dual-path attention (DPA) module suppresses feature redundancy and noise through concurrent channel–spatial attention to enhance discriminative feature representation. The experimental results on hyperspectral image (HSI)–light detection and ranging (LiDAR) and HSI–SAR datasets show that the proposed framework achieves competitive clustering performance and improves multimodal representation learning in most cases.**

***Index Terms*—Attention mechanism, cross-modal information interaction, hyperspectral image (HSI), light detection and ranging (LiDAR) data, semantic association.**

## I. Introduction

THE fusion of multimodal remote sensing (RS) data, such as hyperspectral images (HSIs) and light detection and ranging (LiDAR), has become a pivotal research focus for improving land-cover representation learning and unsupervised scene understanding [1]. HSIs deliver rich spectral information that facilitates material discrimination, whereas LiDAR provides precise geometric features (e.g., elevation and surface structure) [2], [3], [4]. Their complementary integration is essential for capturing comprehensive scene semantics [5]. Nevertheless, achieving effective cross-modal fusion is challenging due to inherent disparities in data characteristics and the complexity of semantic associations across modalities [6], [7], [8].

While deep learning has been widely used for supervised multisource RS analysis [9], developing effective unsupervised clustering frameworks for HSI and LiDAR data remains challenging. Traditional supervised methods require massive labeled samples to maintain accuracy, which leads to high annotation costs and limits their applicability in label-scarce scenarios. In recent years, self-supervised learning has opened up a new path to tackle this challenge. It extracts meaningful feature representations by exploring the inherent attributes and semantic content of large-scale unlabeled data, thus reducing the dependence on expensive labeled data [10]. As a potential solution for unsupervised HSI–LiDAR clustering, self-supervised learning has drawn increasing interest. However, research focusing on its application to multimodal RS clustering remains relatively limited.

Significant progress has been achieved in multimodal HSI–LiDAR fusion for land-cover analysis. Early approaches primarily employed feature-level fusion strategies, which involved simple concatenation of spectral and geometric features before downstream prediction or clustering [10], [11], [12]. Although straightforward, these methods often inadequately captured the complex interdependencies between modalities. Subsequent decision-level fusion frameworks emerged, utilizing separate modality-specific models followed by voting-based or score-level integration [13]. While more flexible, such approaches may incur information loss due to late-stage fusion. The advent of deep learning marked a paradigm shift, as convolutional neural networks (CNNs) demonstrated strong capability in extracting high-level semantic features from multisource data [14]. Subsequent innovations included: encoder–decoder architectures for hierarchical feature learning [15], coupled CNN branches to preserve modality-specific characteristics [16], Gabor-enhanced CNNs for joint spectral–spatial modeling [17]. More recently, cross-attention mechanisms [18] and Transformer-based models [19] have advanced state-of-the-art performance by explicitly modeling long-range dependencies between HSI and LiDAR features [20], [21], [22], [23].

Received 21 April 2026; revised 1 June 2026; accepted 12 June 2026. Date of publication 18 June 2026; date of current version 8 July 2026. This work was supported in part by the Basic Research Program of Jiangsu under Grant BK20252043 and in part by the National Natural Science Foundation of China under Grant 62071168. *(Corresponding author: Hongmin Gao.)*

The authors are with the College of Computer Science and Software Engineering, Hohai University, Nanjing 211100, China (e-mail: xushufang@hhu.edu.cn; 260407010005@hhu.edu.cn; 231607010017@hhu.edu.cn; gaohongmin@hhu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3704974

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

To address these limitations, we propose a multiscale, multigranularity fusion framework comprising four synergistic modules: 1) local patch shuffling (LPS); 2) dynamic group shuffling (DGS); 3) global cross-modality attention (GCMA); and 4) dual-path attention (DPA). The LPS module introduces weak local perturbation in the feature space by applying the same local shuffling order to paired modalities. Therefore, it does not break the spectral–geometric correspondence between co-registered HSI and LiDAR/SAR data, but reduces the model's overdependence on fixed absolute spatial positions and improves robustness to local spatial noise or slight misregistration. The DGS module implements adaptive channel-wise interactions via a data-driven gating network. By dynamically computing interaction weights between semantically meaningful channel groups, this approach eliminates manual rule specification while minimizing cross-modal semantic confusion. The GCMA module establishes comprehensive global interactions through cross-modal attention mechanisms. It computes spatial attention weights across all intermodal position pairs, captures long-range dependencies, and facilitates cross-modal information interaction via multihead attention, thereby enabling the learning of diverse, context-aware semantic associations.

The four modules are designed to address complementary levels of multimodal interaction. LPS works at the local spatial level and serves as a weak regularization mechanism. It jointly shuffles local patches of different modalities using the same permutation, thereby reducing dependence on absolute local positions while retaining the paired spectral–geometric content. DGS works at the channel–group level and learns adaptive groupwise interaction weights, which help exchange semantically related information between modalities without using manually defined fusion rules. GCMA works at the global spatial level and captures long-range cross-modal dependencies that cannot be sufficiently modeled by local or groupwise operations. DPA then refines the fused features through channel–spatial attention, suppressing redundant or noisy responses before the final clustering head. In this way, the hierarchical cross-modal feature shuffling network (HCFSN) forms a progressive fusion process from local perturbation, to groupwise semantic interaction, to global dependency modeling, and finally to discriminative feature refinement.

The main contributions of this article are summarized as follows.

1) We propose HCFSN, an end-to-end unsupervised multimodal clustering framework for HSI–LiDAR/HIS–SAR data. It integrates LPS, DGS, GCMA, and DPA into a hierarchical fusion pipeline.
2) The proposed LPS and DGS modules model local robustness and channel–group interaction, respectively. LPS applies shared weak perturbation to paired multimodal patches without destroying their physical correspondence, while DGS adaptively exchanges information between semantically related channel groups.
3) The GCMA and DPA modules further enhance cross-modal representation learning from global and refinement levels. GCMA captures long-range cross-modal dependencies, and DPA suppresses redundant channel–spatial responses before clustering.
4) Extensive experiments on Trento, MUUFL, and Berlin demonstrate that HCFSN achieves competitive performance on most major clustering metrics. Additional ablation, feature separability, and complexity analyses further verify the effectiveness and computational feasibility of the proposed framework.

## II. Related Works

### A. Contrastive Learning

Contrastive learning, a prominent paradigm in self-supervised learning, learns discriminative feature representations by optimizing instance relationships: maximizing agreement between semantically similar samples (positive pairs) while minimizing similarity between unrelated instances (negative pairs) [24], [25]. This approach significantly reduces dependence on labeled data by leveraging intrinsic structural and semantic relationships within unannotated datasets, demonstrating particular value for RS applications where manual annotations are often limited [26].

Contrastive learning operates through carefully designed pretext tasks that compel models to extract semantically meaningful patterns. The widely adopted InfoNCE loss function formalizes this approach by approximating the mutual information lower bound between anchor–positive sample pairs, while simultaneously enforcing discrimination against negative samples. This framework comprises three core components as follows.

1) *Data Augmentation:* Constructs positive pairs by generating multiple views of each instance through transformations (e.g., random cropping, flipping, or spectral perturbations for RS data [27]), preserving semantic invariance across augmentations.
2) *Feature Encoding:* Implements deep neural architectures (CNNs or Transformers) to project raw inputs into a latent embedding space that maintains semantic relationships.
3) *Similarity Measurement:* Utilizes distance metrics (cosine similarity or Euclidean distance) to quantify feature relationships, optimized through contrastive objectives (InfoNCE, NT-Xent) that maximize agreement between positive pairs and dispersion of negatives.

Early contrastive learning approaches like momentum contrast (MoCo) circumvented the computational demands of large batch sizes by implementing a dynamic memory bank for efficient negative sample storage and retrieval. Subsequent advancements, including Bootstrap Your Own Latent (BYOL) [28] and Simple Siamese (SimSiam) [29], demonstrated that negative samples could be eliminated through innovative architectural designs—utilizing asymmetric networks and stop-gradient operations to prevent representation collapse. Barlow Twins [30] further streamlined the framework by introducing a redundancy reduction objective that achieves competitive performance even with limited training data.

In RS applications, contrastive learning has proven particularly effective for hyperspectral (HS) image analysis and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

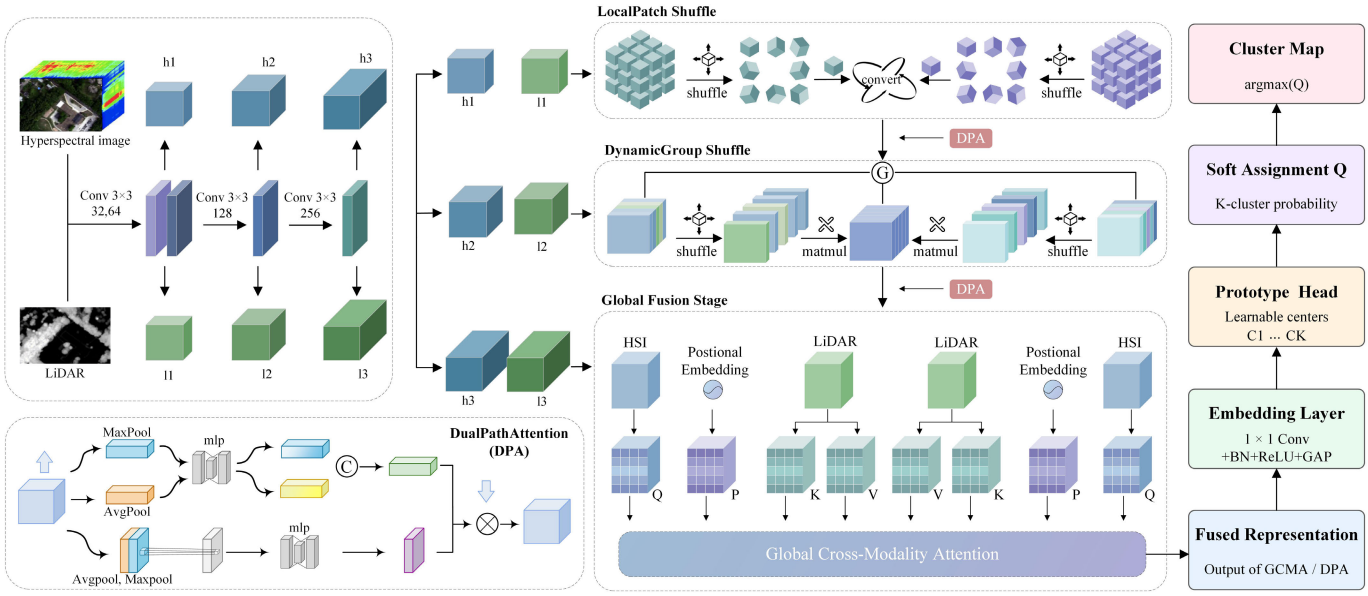



Fig. 1. Overall structure of the HCFSN framework. The framework consists of four modules: the LPS module, the DGS module, the GCMA module, and the DPA module. It processes from local patch features to spectral features and then to global features.

multimodal fusion tasks. Contrastive clustering [31] enhanced feature separability by jointly optimizing contrastive and clustering objectives for single-modality RS data. Recent multimodal extensions adapt this paradigm to cross-modal scenarios, establishing shared latent spaces that align complementary features from heterogeneous sensors (e.g., HSI, LiDAR, and SAR) while preserving modality-specific characteristics.

### B. Multimodal Clustering

Multimodal clustering aims to group data from diverse sources such as images, text, and audio by leveraging their underlying correlations [32], [33]. Self-supervised learning has shown promise in extracting semantically consistent representations from multimodal data, facilitating the clustering process by capturing intermodal similarities.

Multiview subspace clustering (MVSC) [34] is a common framework that projects multimodal data into a shared subspace. However, traditional MVSC methods suffer from high computational costs on large-scale datasets. Recent scalable extensions, such as LMVSC, EOMSC-CA, FPMVS-CAG, and SMVSC, employ anchor-based strategies to reduce complexity. While efficient, these methods often rely on linear or shallow models, limiting their ability to handle complex nonlinear multimodal relationships.

Deep learning approaches like deep CCA (DCCA) and autoencoders (AEs) can learn more expressive representations. However, many deep clustering methods—such as AE networks (AE2-Nets) and multisensor deep clustering (MDC)—adopt a two-stage process, where feature learning and clustering are decoupled. This separation often leads to suboptimal performance [35], [36]. In contrast, our approach unifies adaptive modal fusion, discriminative representation learning, and online clustering within an end-to-end trainable model.

## III. Methodology

As shown in Fig. 1, HCFSN is designed as an end-to-end unsupervised multimodal clustering framework. Given two modalities, i.e., HSI and LiDAR/SAR, two augmented views are first generated for each multimodal sample. The augmented samples are then fed into a modality-specific encoder and a hierarchical cross-modal interaction module, which progressively performs local patch-level interaction, channel–group interaction, and global cross-modal attention. The fused high-level feature is further mapped into a compact embedding by an embedding layer. Instead of applying an external clustering algorithm after feature extraction, HCFSN employs a prototype-based clustering head to directly produce soft cluster assignments. The final cluster label of each sample is obtained by selecting the cluster with the maximum assignment probability. The whole network, including the feature encoder, cross-modal interaction modules, embedding layer, and clustering head, is optimized jointly using an assignment-level contrastive loss and a clustering loss.

The four modules are organized according to a progressive local-to-global fusion rationale rather than being used as independent components. LPS first introduces a weak local perturbation with a shared shuffling order across modalities, which reduces overreliance on fixed spatial layouts while preserving paired spectral–geometric content. DGS then performs adaptive channel–group interaction on the locally regularized features, enabling semantically related groups from different modalities to exchange information. GCMA further extends the interaction from local and channel–group levels to the global spatial context by modeling long-range cross-modal dependencies. Finally, DPA refines the fused representation by jointly emphasizing informative channels and discriminative spatial regions before the clustering head. Therefore, HCFSN follows a hierarchical design from local robustness, to group-wise semantic exchange, to global dependency modeling, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

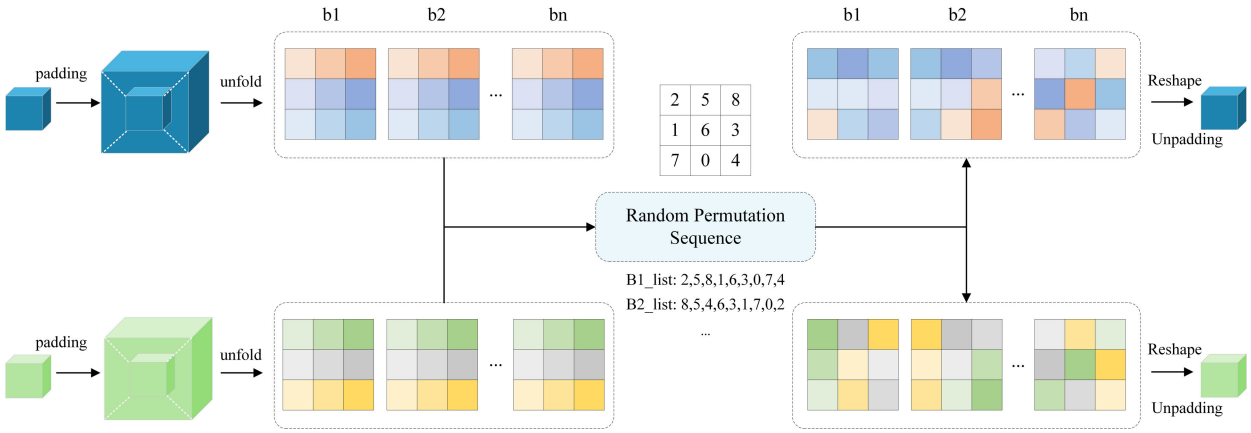



Fig. 2. LPS module.

finally to discriminative feature refinement for unsupervised clustering.

### A. LPS Module

HSI and LiDAR/SAR data are usually co-registered, and reliable spatial correspondence is an important basis for multimodal fusion, as also emphasized in recent cross-modal RS matching studies [37]. Therefore, the purpose of LPS is not to destroy the physical correspondence between modalities. Instead, LPS applies the same local shuffling order to both modalities in the feature space. In this way, the paired multimodal content is preserved, while the network is prevented from relying only on fixed absolute spatial positions. This weak local perturbation encourages the model to learn more robust semantic associations within local neighborhoods and provides locally regularized inputs for the following channel–group and global cross-modal interaction modules. As illustrated in Fig. 2, the same permutation index is used for the paired local patches from different modalities. Thus, each shuffled HSI patch is still matched with its corresponding LiDAR/SAR patch, and only the local ordering of patch units is perturbed. This design differs from independent random shuffling, which would destroy cross-modal correspondence and introduce physically meaningless pairs.

Taking the feature maps of HSI and LiDAR, we obtain $\mathbf{F_H} \in R^{C\times H\times W}$ and $\mathbf{F_L} \in R^{C\times H\times W}$ after a simple convolution process. First, we perform an auto-padding process for $\mathbf{F_H}$ and $\mathbf{F_L}$ to ensure that the height and width can be divisible by the patch_size, and then, we take the padded feature maps $\mathbf{F_{ph}} \in R^{C\times(H+pad_h)\times(W+pad_w)}$ and $\mathbf{F_{pl}} \in R^{C\times(H+pad_h)\times(W+pad_w)}$ expanded into p × p localized blocks, i.e., $\mathbf{F'_{ph}} \in R^{C\times(H+pad_h//p)\times(W+pad_w//p)\times p\times p}$ and $\mathbf{F'_{pl}} \in R^{C\times(H+pad_h//p)\times(W+pad_w//p)\times p\times p}$

$$\mathbf{F'_{ph}} = Unfold\left(Pad\left(\mathbf{F_H}\right)\right) \tag{1}$$

$$\mathbf{F'_{pl}} = Unfold\left(Pad\left(\mathbf{F_L}\right)\right). \tag{2}$$

Then, generate a different randomized arrangement order for each batch, calculate the total number of blocks (H + pad_h)//p * (W + pad_w)//p to generate B randomized arrangement perm in the shape of [B, num_patches]. Reorganize the local blocks according to the random permutation index by first reshape the $\mathbf{F'_{ph}}$ and $\mathbf{F'_{pl}}$ as [B, C, num_patches, p, p] then use gather to rearrange them according to the perm's index to obtain $\mathbf{S_h} \in R^{C\times num_patches\times P\times P}$ and $\mathbf{S_l} \in R^{C\times num_patches\times P\times P}$

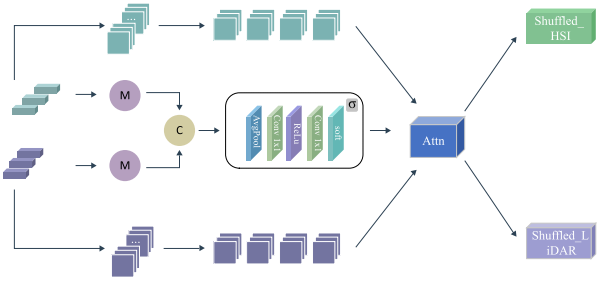



Fig. 3. DGS module.

$$\mathbf{S_h} = Gather\left(\mathbf{F''_{ph}}, perm\right) \tag{3}$$

$$\mathbf{S_l} = Gather\left(\mathbf{F''_{pl}}, perm\right). \tag{4}$$

Finally, the disrupted partial blocks are reassembled into a complete feature map by first reshaping as [B, C, H//p, W//p, p, p], adjusting the dimensional order using permute, and finally reshaping as the original shape [B, C, H + pad_h, W + pad_w], removing the padding part, and returning the original dimensions $\mathbf{F_c} \in R^{C\times H\times W}$ and $\mathbf{F_s} \in R^{C\times H\times W}$

$$\mathbf{F_c} = Unpad\left(Reshape\left(permute\left(\mathbf{S_h}\right)\right)\right) \tag{5}$$

$$\mathbf{F_s} = Unpad\left(Reshape\left(Permute\left(\mathbf{S_l}\right)\right)\right). \tag{6}$$

### B. DGS Module

HSI and LiDAR/SAR features inherently encode distinct semantic information through their channel dimensions, where HSI channels represent spectral responses and LiDAR/SAR channels provide geometric or backscattering-related cues. Simple feature concatenation or elementwise summation may create semantic conflicts between these heterogeneous representations. As shown in Fig. 3, the DGS module implements adaptive channel–group interactions by: 1) dividing channels into several groups; 2) computing dynamic affinity weights

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

between cross-modal groups; and 3) selectively exchanging information between strongly correlated groups. This data-driven design reduces the need for manually specified interaction rules while preserving modality-specific characteristics.

First, the cross-modal group interaction weights are dynamically generated by a gating network. Given the shuffled HSI feature $\mathbf{F}_c \in \mathbb{R}^{B\times C\times H\times W}$ and the shuffled LiDAR/SAR feature $\mathbf{F}_s \in \mathbb{R}^{B\times C\times H\times W}$, global average pooling is first applied to each modality to obtain two channel descriptors. The two descriptors are concatenated and fed into a lightweight gating network consisting of two $1 \times 1$ convolutional layers, ReLU activation, and a softmax function

$$\mathbf{A} = \text{Gate}\left(\text{Concat}\left(\text{GAP}\left(\mathbf{F}_c\right), \text{GAP}\left(\mathbf{F}_s\right)\right)\right) \tag{7}$$

where $\mathbf{A} \in \mathbb{R}^{B\times G\times G}$ denotes the dynamic group interaction matrix, and $G$ is the number of channel groups.

To perform groupwise interaction, the channel dimension is divided into $G$ groups. Let $d = C/G$ be the number of channels in each group. The two modal features are reshaped as

$$\mathbf{F}_c^g, \mathbf{F}_s^g \in \mathbb{R}^{B\times G\times d\times H\times W}. \tag{8}$$

The interaction matrix $\mathbf{A}$ is then used to aggregate cross-modal group information. Specifically, the HSI-side updated feature is obtained by weighting LiDAR/SAR groups, while the LiDAR/SAR-side updated feature is obtained by the transposed interaction direction

$$\widetilde{\mathbf{F}}_c^f = \sum_{g=1}^{G} \mathbf{A}_{f,g}\mathbf{F}_s^g, \quad f = 1, \ldots, G \tag{9}$$

$$\widetilde{\mathbf{F}}_s^f = \sum_{g=1}^{G} \mathbf{A}_{g,f}\mathbf{F}_c^g, \quad f = 1, \ldots, G. \tag{10}$$

Finally, the grouped features are reshaped back to $\widetilde{\mathbf{F}}_c, \widetilde{\mathbf{F}}_s \in \mathbb{R}^{B\times C\times H\times W}$. In this way, DGS performs adaptive channel–group exchange between modalities rather than using fixed manually defined interaction rules.

### C. Global Cross-Modality Attention

To comprehensively model cross-modal dependencies between HS (HSI) and LiDAR data, we propose the GCMA module. This architecture facilitates adaptive feature interaction while maintaining spatial contextual integrity through three key mechanisms: 1) cross-modal attention weights that dynamically capture discriminative spectral–geometric correlations; 2) cross-modal information interaction preserving modality-specific characteristics; and 3) hierarchical feature aggregation that overcomes the limitations of static fusion approaches [38], [39], [40].

The GCMA module consists of query/key/value projection layers, positional encoding, and residual cross-modal fusion, as shown in Fig. 4. Given an HSI feature $\mathbf{F}_H \in \mathbb{R}^{B\times C\times H\times W}$ and a LiDAR/SAR feature $\mathbf{F}_L \in \mathbb{R}^{B\times C\times H\times W}$, GCMA first models the HSI-to-LiDAR/SAR attention direction. The query is generated from the HSI feature, while the key and value are generated from the LiDAR/SAR feature

$$\mathbf{Q}_H = \phi_q\left(\mathbf{F}_H\right) + \mathbf{PE}$$

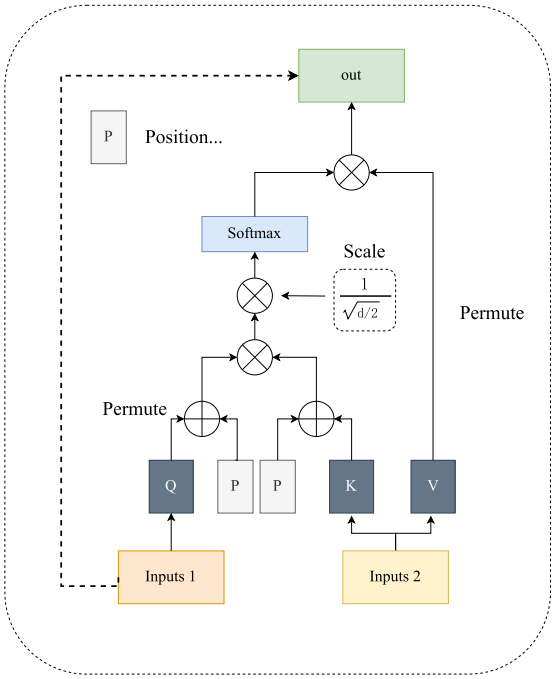



Fig. 4. GCMA module.

$$\mathbf{K}_L = \phi_k\left(\mathbf{F}_L\right) + \mathbf{PE}$$
$$\mathbf{V}_L = \phi_v\left(\mathbf{F}_L\right) \tag{11}$$

where $\phi_q(\cdot)$ and $\phi_k(\cdot)$ are $1 \times 1$ convolutional projections that reduce the channel dimension from $C$ to $C/2$, and $\phi_v(\cdot)$ preserves the channel dimension. Therefore, $\mathbf{Q}_H, \mathbf{K}_L \in \mathbb{R}^{B\times(C/2)\times H\times W}$ and $\mathbf{V}_L \in \mathbb{R}^{B\times C\times H\times W}$. $\mathbf{PE} \in \mathbb{R}^{1\times(C/2)\times H\times W}$ denotes the learnable positional encoding.

The projected tensors are reshaped into sequence format

$$\mathbf{Q}_H \in \mathbb{R}^{B\times HW\times(C/2)}$$
$$\mathbf{K}_L \in \mathbb{R}^{B\times(C/2)\times HW}$$
$$\mathbf{V}_L \in \mathbb{R}^{B\times C\times HW}. \tag{12}$$

The cross-modal attention matrix is computed as

$$\mathbf{M}_{H\leftarrow L} = \text{Softmax}\left(\frac{\mathbf{Q}_H\mathbf{K}_L}{\sqrt{C/2}}\right) \tag{13}$$

where $\mathbf{M}_{H\leftarrow L} \in \mathbb{R}^{B\times HW\times HW}$ measures the spatial correspondence from LiDAR/SAR features to HSI query positions. The attended LiDAR/SAR feature is then obtained by

$$\mathbf{V}_{H\leftarrow L} = \mathbf{V}_L\mathbf{M}_{H\leftarrow L}^T \tag{14}$$

and reshaped back to $\mathbb{R}^{B\times C\times H\times W}$. The HSI-side global fusion output is computed by a residual connection

$$\mathbf{F}_{H\leftarrow L} = \mathbf{F}_H + \gamma\mathbf{V}_{H\leftarrow L} \tag{15}$$

where $\gamma$ is a learnable scalar controlling the strength of cross-modal fusion.

In the implementation, GCMA is applied in two directions by swapping the query and key/value modalities. Thus, the LiDAR/SAR-side output $\mathbf{F}_{L\leftarrow H}$ is obtained in the same way by using LiDAR/SAR as the query and HSI as the key/value. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

two directional outputs are then refined by the DPA module and combined to form the global fused representation

$$\mathbf{F}_G = \mathrm{DPA}\left(\mathbf{F}_{H\leftarrow L}\right) + \mathrm{DPA}\left(\mathbf{F}_{L\leftarrow H}\right). \tag{16}$$

This bidirectional formulation makes the direction of cross-modal attention explicit and preserves complementary information from both modalities.

### D. Dual-Path Attention

To enhance discriminative feature representation while suppressing noise in multimodal RS data, we propose the DPA. This architecture synergistically integrates channel-wise and spatial-wise attention mechanisms to adaptively amplify informative spectral channels and salient spatial regions, effectively addressing the prevalent challenges of feature redundancy and irrelevant activations in fused HSI–LiDAR representations [41]. The DPA module employs two parallel computational pathways: 1) a channel attention branch that models spectral channel interdependencies and 2) a spatial attention branch that captures contextual relationships across spatial locations. Through their joint operation, these complementary attention mechanisms progressively refine feature maps by simultaneously emphasizing both spectrally significant and spatially relevant features.

Channel attention focuses on weighting spectral channels (for HSI) or geometric attributes (for LiDAR) based on their discriminative power. First, use AdaptiveAvgPool2d (1) to compress spatial dimensions into a channel-wise statistic vector, capturing global spectral/spatial characteristics

$$\mathbf{A_f} = GlobalAvgPool\left(\mathbf{R_x}\right) \tag{17}$$

where $\mathbf{A_f} \in R^{B\times C\times 1\times 1}$ and $\mathbf{R_x}$ are the HSI and LiDAR features obtained after processing in each module.

Then, a lightweight convolution network ($1\times 1$ conv layers) with ReLU activation learns channel dependencies, followed by a Sigmoid function to generate channel weights

$$\mathbf{CA_x} = Sigmoid\left(Conv\left(ReLU\left(Conv\left(\mathbf{A_f}\right)\right)\right)\right) \tag{18}$$

where $\mathbf{CA_x} \in R^{B\times C\times 1\times 1}$ is the channel weight obtained from the channel attention function.

Spatial attention focuses on modeling local spatial dependencies, highlighting discriminative regions (e.g., object boundaries or homogeneous areas) while suppressing noisy or irrelevant spatial locations. This is critical for RS data, where spatial context (e.g., the shape of buildings or continuity of roads) plays a key role in semantic understanding [42].

First, compute two spatial statistic maps $\mathbf{m_feat}$ and $\mathbf{a_feat}$ by aggregating channel-wise information, where $\mathbf{m_feat} \in R^{B\times 1\times H\times W}$ capturing local contrast, $\mathbf{a_feat} \in R^{B\times 1\times H\times W}$ capturing regional homogeneity.

Next, combine the two statistic maps to preserve complementary spatial information $\mathbf{c_feat} \in R^{B\times 2\times H\times W}$.

A $3\times 3$ convolutional layer with padding = 1 is applied to $\mathbf{c_feat}$ to model local spatial correlations, followed by a Sigmoid activation to generate spatial attention weights

$$\mathbf{SA_x} = Sigmoid\left(Conv\left(\mathbf{c_feat}\right)\right) \tag{19}$$

where $\mathbf{SA_x} \in R^{B\times 1\times H\times W}$ is the spatial weight obtained from the spatial attention function.

These weights adaptively enhance regions with strong semantic relevance (e.g., the contour of a road in LiDAR data) and weaken noisy areas (e.g., sparse LiDAR points). The final output of the DPA module is obtained by multiplying the input feature map with both channel and spatial attention weights

$$\mathbf{O_x} = \mathbf{R_x} \odot \mathbf{CA_x} \odot \mathbf{SA_x}. \tag{20}$$

### E. Unsupervised Clustering Head and Joint Objective

After hierarchical cross-modal fusion, the last-level fused feature is passed through an embedding layer consisting of a $1\times 1$ convolution, batch normalization, ReLU activation, and global average pooling. This produces a compact embedding for each multimodal sample. Let $h_i$ be the embedding of the $i$th sample. The clustering head maintains $K$ learnable cluster centers, where $K$ is set to the number of land-cover categories in each dataset. The soft assignment between $h_i$ and the $k$th cluster center is computed according to a Student's t-distribution kernel. Specifically, samples closer to a cluster center obtain higher assignment probabilities, and all assignment scores are normalized over $K$ clusters. The predicted cluster label is then obtained by the maximum assignment probability.

The cluster centers are randomly initialized with Xavier normal initialization and are treated as learnable parameters. During training, they are updated by backpropagation together with the encoder, the cross-modal interaction modules, and the embedding layer. To allow the clustering head to adapt more rapidly to the evolving representation space, the cluster centers are optimized with a scaled learning rate. No ground-truth labels are used for initializing or updating the cluster centers.

For each sample, two augmented multimodal views are generated and passed through the same network, producing two soft assignment distributions. An assignment-level contrastive loss is applied to these two distributions. This loss encourages the two augmented views of the same sample to have consistent cluster assignments while pushing apart the assignment distributions of different samples within the same mini-batch. Therefore, the contrastive term does not simply regularize low-level features, but directly improves the discriminability and consistency of the clustering-oriented assignment space.

In addition to the assignment-level contrastive loss, a clustering loss is imposed to refine the soft assignments. Following the self-training principle in deep clustering, a sharpened target distribution is constructed from the current assignment probabilities. The clustering loss minimizes the divergence between the predicted assignment distribution and the target distribution, thereby encouraging more confident and stable cluster assignments. Furthermore, an assignment-balance regularization term is used to reduce degenerate solutions in which most samples are assigned to only a few clusters. A center regularization term is also applied to encourage the cluster centers to remain discriminative in the embedding space.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DISTRIBUTION OF SAMPLES FOR THE TRENTO DATASET

| No. | Class Name | Samples |
|---|---|---|
| 1 | Apple trees | 4034 |
| 2 | Buildings | 2903 |
| 3 | Ground | 479 |
| 4 | Woods | 9123 |
| 5 | Vineyard | 10501 |
| 6 | Roads | 3174 |
| | Total | 30214 |

TABLE II
DISTRIBUTION OF SAMPLES FOR THE MUUFL DATASET

| No. | Class Name | Samples |
|---|---|---|
| 1 | Trees | 23246 |
| 2 | Mostly grass | 4270 |
| 3 | Mixed ground surface | 6882 |
| 4 | Dirt and sand | 1826 |
| 5 | Road | 6687 |
| 6 | Water | 466 |
| 7 | Building shadow | 2233 |
| 8 | Building | 6240 |
| 9 | Sidewalk | 1385 |
| 10 | Yellow curb | 183 |
| 11 | Cloth panels | 269 |
| | Total | 53687 |

The final objective is composed of the assignment-level contrastive loss, the clustering loss, and the regularization terms. The contrastive loss improves the consistency and separability of the assignment distributions under different augmentations, while the clustering loss directly optimizes the prototype-based assignment structure. These two objectives are complementary: the former enhances representation robustness, and the latter aligns the learned representation with the unsupervised clustering objective.

## IV. EXPERIMENTAL RESULTS AND ANALYSIS

### A. Datasets and Evaluation Metric

*1) HS–LiDAR Trento Dataset:* The dataset was acquired over a rural region south of Trento, Italy, using the AISA Eagle HS sensor, with co-registered HS and LiDAR data at 1-m spatial resolution (600 × 166 pixels). The HS data consists of 63 spectral bands covering the visible to near-infrared range (0.40–0.98 $\mu$m), while the LiDAR data comprises two channels (elevation and intensity). The dataset encompasses six distinct land-cover classes, with the sample distribution for each class detailed in Table I.

*2) HS–LiDAR MUUFL Dataset:* The dataset was acquired in November 2010 at the University of Southern Mississippi's Gulfport campus (Long Beach, Mississippi, USA), comprising co-registered HSI and LiDAR-derived digital surface models (DSMs) at 325 × 220 pixel resolution. The HSI data were collected using the ITRES Compact Airborne Spectral Imager (CASI-1500) with 64 spectral channels spanning the visible to near-infrared spectrum (0.38–1.05 $\mu$m) at 0.54 × 1.0 m spatial resolution. The LiDAR data was acquired by an ALTM sensor (1064-nm wavelength) with 0.60 × 0.78 m spatial resolution. The land-cover clustering scheme and corresponding sample counts are detailed in Table II.

TABLE III
DISTRIBUTION OF SAMPLES FOR THE BERLIN DATASET

| No. | Class Name | Samples |
|---|---|---|
| 1 | Forest | 54954 |
| 2 | Residential Area | 268642 |
| 3 | Industrial Area | 19566 |
| 4 | Low Plants | 59282 |
| 5 | Soil | 17426 |
| 6 | Allotment | 13305 |
| 7 | Commercial Area | 24824 |
| 8 | Water | 6672 |
| | Total | 464671 |

*3) HS–SAR Berlin Dataset:* The dataset comprises simulated EnMAP imagery synthesized from HyMap HS data, covering urban and adjacent rural areas of Berlin at 30-m ground sampling distance (GSD). This freely available dataset (797 × 220 pixels) contains 244 spectral bands spanning the visible to shortwave infrared spectrum (0.4–2.5 $\mu$m). The land-cover clustering scheme and corresponding sample distributions are detailed in Table III.

The clustering performance is quantitatively evaluated using five established metrics: 1) overall accuracy (OA), calculated as the ratio of correctly predicted samples to total samples; 2) kappa coefficient, measuring clustering agreement beyond chance expectation; 3) normalized mutual information (NMI), quantifying the shared information between predicted and ground-truth clusters; 4) adjusted rand index (ARI), assessing the similarity between cluster assignments; and 5) purity, indicating the degree of class homogeneity within clusters, where higher values reflect better performance. Following standard evaluation protocols, predicted cluster labels are first aligned with ground-truth labels using the Hungarian algorithm [43] before metric computation.

### B. Implementation Details

The model is trained on 7 × 7 spatial blocks extracted from each modality as input samples. We employ the Adam optimizer with an initial learning rate of $5 \times 10^{-4}$, which is decayed by a factor of 0.1 every ten epochs. Training proceeds for 100 epochs using a batch size of 256, with $\ell_2$ regularization (weight decay = $5 \times 10^{-4}$). Unless otherwise specified, the loss-balancing coefficient $\lambda$ and the temperature coefficient $\tau$ are fixed at 1.0 and 0.5, respectively. All implementations utilize PyTorch and are executed on an NVIDIA RTX 3090 GPU with 24 GB memory.

To maintain an unsupervised training and evaluation protocol, ground-truth labels are not used for representation learning, cluster-center updating, loss optimization, hyperparameter selection, model selection, or early stopping. The proposed model is trained under predefined settings with a fixed number of epochs, and the final checkpoint is used for evaluation. The number of clusters is set according to the known number of land-cover categories in each benchmark dataset, which is a common setting in unsupervised clustering

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

evaluation. After training, each sample is assigned to the cluster with the maximum assignment probability. Ground-truth labels are used only after clustering, where the predicted cluster indices are aligned with the reference labels using the Hungarian algorithm before computing OA, kappa, NMI, ARI, purity, and AA.

For baseline comparison, we follow a unified evaluation protocol while keeping the method-specific input settings of different baselines. Since the compared methods are designed under different technical assumptions and input forms, forcing all of them to use exactly the same patch size or dimensionality-reduction setting may lead to unfair or suboptimal results. Therefore, we use publicly available implementations released by the authors whenever available and keep their main model structures unchanged. For methods that require patch-based inputs, suitable patch settings are adopted according to their original implementations and input requirements. For methods that require vectorized features or dimensionality reduction, the corresponding multimodal features are used, and PCA is applied when required by the method or recommended by the original implementation. For all compared methods, the same datasets, sample masks, number of clusters, label-matching strategy, and metric computation procedure are used.

For HCFSN, the same main hyperparameter settings are used across all datasets unless otherwise specified. The patch size, batch size, learning rate, weight decay, number of training epochs, loss-balancing coefficient $\lambda$, and temperature coefficient $\tau$ are predefined before the main comparison experiments. Specifically, $\lambda$ is set to 1.0, and $\tau$ is set to 0.5 according to the parameter sensitivity analysis and the stable training behavior observed in preliminary experiments. These settings are then kept unchanged on the Trento, MUUFL, and Berlin datasets. The learning rate follows a step-decay schedule, where it is reduced by a factor of 0.1 every ten epochs. The stopping criterion is the fixed training epoch, and the final checkpoint is used for evaluation without label-based model selection.

For clarity, all compared methods are described as existing baseline methods, and the term "proposed method" is used only to refer to HCFSN in this article.

1) *LMVSC:*[1] LMVSC is a scalable MVSC method that uses anchor-based representations to reduce the computational cost of large-scale multiview clustering.
2) *EOMSC-CA:*[2] EOMSC-CA is an orthogonal MVSC method with a channel-attention mechanism.
3) *BMVC:*[3] BMVC is a bilinear multiview clustering method that models second-order feature interactions for multiview representation learning.
4) *FPMVS-CAG:*[4] FPMVS-CAG is a feature-preserving MVSC method that integrates channel attention with graph-structured representation learning.
5) *SMVSC:*[5] SMVSC is a self-supervised MVSC method that learns representations from heterogeneous data sources without manual labels.
6) *DCCA:*[6] DCCA is a deep canonical correlation analysis method that learns nonlinear cross-modal correlations through joint representation learning.
7) *AE2-Nets:*[7] AE2-Nets is an AE-based multiview representation learning method that uses encoder–decoder reconstruction to learn a shared latent representation for clustering.
8) *MDC:*[8] MDC is a multimodal deep clustering method that combines deep representation learning with clustering-oriented optimization.
9) *TMPCC:*[9] TMPCC is a Transformer-based contrastive prototypical clustering method for multimodal RS data.

### C. Clustering Accuracy and Discussion

Our experimental evaluation compares HCFSN against several representative clustering methods on the Trento, MUUFL, and Berlin datasets. The quantitative results in Tables IV–VI show that HCFSN achieves competitive and generally strong clustering performance across different scenes. Specifically, HCFSN obtains the best OA on the Trento and MUUFL datasets, reaching 94.53% and 68.05%, respectively. On the Berlin dataset, HCFSN achieves the best kappa, NMI, purity, and AA, but its OA is lower than that of the best competing method. These results indicate that HCFSN improves overall clustering consistency and cross-modal representation learning in most cases. Meanwhile, the relatively low AA on MUUFL and the nonbest OA on Berlin suggest that HCFSN still faces challenges in minority-class clustering and cross-sensor generalization, especially for HSI–SAR data.

Table IV summarizes the clustering performance on the Trento dataset, with the corresponding visualization shown in Fig. 5. HCFSN achieves the best results on the major evaluation metrics, including OA, AA, and kappa, indicating that the proposed hierarchical interaction design is effective for HSI–LiDAR clustering. Notably, the ground class shows relatively lower clustering accuracy, which is also observed in several comparative methods. This may be related to its small sample proportion, accounting for only 1.59% of the total samples, and the difficulty of discriminating natural ground surfaces due to spectral variability and textural homogeneity. These results support the effectiveness of HCFSN on Trento, while also showing that small or visually ambiguous land-cover categories remain challenging in unsupervised clustering.

Table V presents the clustering performance on the MUUFL dataset, with the visual results shown in Fig. 6. HCFSN achieves the best ACC, kappa, and ARI, reaching 68.05%, 58.32%, and 54.55%, respectively. These results indicate that HCFSN can learn effective multimodal representations at the

[1]https://github.com/sckangz/LMVSC
[2]https://github.com/Tracesource/EOMSC-CA
[3]https://github.com/DarrenZZhang/TPAMI19_BMVC
[4]https://github.com/wangsiwei2010/FPMVS-CAG
[5]https://github.com/Jeaninezpp/SMVSC
[6]https://github.com/Michaelvll/DeepCCA
[7]https://github.com/AkChen/AE2-Nets
[8]https://github.com/Kasra2020/MDC
[9]https://github.com/AngryCai/TMPCC

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CLUSTERING PERFORMANCE (%) ON THE TRENTO DATASET. THE BEST RESULTS ARE MARKED IN BOLD

| Class | LMVSC | EOMSC-CA | BMVC | FPMVS-CAG | SMVSC | DCCA | AE2Net | MDC | TMPCC | Ours |
|---|---|---|---|---|---|---|---|---|---|---|
| Apple trees | 98.51 | 0.37 | 80.02 | 58.83 | 59.17 | 74.86 | 78.41 | 93.28 | 79.13 | **99.93** |
| Buildings | 81.85 | 24.70 | 61.97 | 2.10 | 4.58 | 47.26 | 66.59 | 99.41 | **100.00** | 89.11 |
| Ground | 0.00 | 0.00 | 0.84 | **10.65** | 21.71 | 0.00 | 1.04 | 0.00 | 2.09 | 0.00 |
| Woods | 98.14 | 96.01 | 96.84 | 93.72 | 92.57 | 99.50 | 43.97 | 98.48 | **100.00** | **100.00** |
| Vineyard | 49.63 | 99.55 | 51.70 | 79.50 | 74.71 | 88.74 | 55.32 | 64.75 | 98.33 | **100.00** |
| Roads | 62.35 | 0.00 | 49.91 | 92.53 | 91.93 | 67.33 | 75.14 | 85.95 | 79.68 | 73.06 |
| ACC | 74.45 | 72.30 | 69.10 | 73.87 | 72.26 | 82.49 | 57.28 | 83.27 | 92.95 | **94.53** |
| Kappa | 67.43 | 63.71 | 60.71 | 65.01 | 63.63 | 76.27 | 46.57 | 78.40 | 90.63 | **92.69** |
| NMI | 68.85 | 61.79 | 61.44 | 62.22 | 61.81 | 67.58 | 47.82 | 76.82 | 90.56 | **91.08** |
| ARI | 60.92 | 46.35 | 61.79 | 61.27 | 61.78 | 70.36 | 34.67 | 70.09 | 94.22 | **95.54** |
| Purity | 79.30 | 66.75 | 74.22 | 73.93 | 74.14 | 83.16 | 68.67 | 83.94 | **95.59** | 94.53 |
| AA | 65.08 | 36.77 | 57.67 | 56.22 | 55.35 | 62.95 | 53.41 | 73.64 | 76.54 | **77.02** |
| Main Time(s) | 1366.97 | 3722.84 | 837.31 | 321.26 | 863.39 | 9057.49 | 556.27 | 253.32 | 1493.95 | 1028.21 |

TABLE V
CLUSTERING PERFORMANCE (%) COMPARISON ON THE MUUFL DATASET. THE BEST RESULTS ARE MARKED IN BOLD

| Class | LMVSC | EOMSC-CA | BMVC | FPMVS-CAG | SMVSC | DCCA | AE2Net | MDC | TMPCC | Ours |
|---|---|---|---|---|---|---|---|---|---|---|
| Trees | 43.42 | 45.63 | 29.95 | 52.51 | 55.17 | 70.07 | 31.59 | 24.40 | 52.50 | **79.31** |
| Mostly grass | 84.99 | 0.54 | 61.24 | 76.07 | **94.22** | 45.97 | 9.77 | 69.86 | 82.79 | 0.00 |
| Mixed ground surface | 64.40 | **93.20** | 39.64 | 56.79 | 13.22 | 60.24 | 61.74 | 51.57 | 34.42 | 80.43 |
| Dirt and sand | 9.26 | 10.35 | 69.77 | 0.00 | 1.37 | 4.60 | **84.99** | 14.46 | 1.53 | 0.00 |
| Road | 91.19 | 7.12 | 71.78 | 63.64 | 68.09 | 78.26 | 77.61 | 84.58 | 91.27 | **96.26** |
| Water | 0.00 | 0.00 | **12.66** | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 10.09 | 0.00 |
| Building shadow | **78.28** | 28.97 | 64.76 | 0.22 | 0.13 | 0.90 | 61.71 | 73.18 | 65.47 | 0.00 |
| Building | 40.50 | 88.50 | 46.17 | 13.17 | 15.38 | 45.51 | 50.21 | 61.52 | 92.55 | **98.21** |
| Sidewalk | 0.14 | 25.14 | **43.54** | 0.22 | 2.02 | 0.00 | 0.29 | 1.88 | 2.74 | 0.00 |
| Yellow curb | 15.30 | 1.64 | 0.00 | 10.93 | **25.14** | 0.00 | 6.56 | 0.00 | 0.00 | 0.00 |
| Cloth panels | **66.91** | 34.29 | 0.37 | 0.00 | 0.00 | 0.00 | 40.89 | 0.00 | 0.00 | 0.00 |
| ACC | 53.84 | 45.31 | 43.53 | 45.42 | 44.33 | 56.95 | 43.56 | 44.00 | 58.79 | **68.05** |
| Kappa | 45.75 | 30.70 | 37.04 | 30.48 | 28.85 | 44.51 | 34.96 | 37.04 | 51.06 | **58.32** |
| NMI | 52.81 | 30.91 | 47.30 | 39.10 | 35.12 | 37.44 | 42.87 | 46.52 | **59.02** | 50.49 |
| ARI | 38.65 | 17.24 | 27.24 | 25.22 | 22.48 | 35.94 | 23.49 | 22.96 | 45.33 | **54.55** |
| Purity | 75.89 | 48.27 | 73.35 | 59.81 | 54.27 | 60.90 | 65.75 | 70.57 | **77.80** | 68.05 |
| AA | **44.94** | 30.49 | 39.99 | 24.87 | 25.52 | 27.78 | 38.67 | 34.68 | 39.40 | 32.20 |
| Time (s) | 2555.94 | 253.85 | 222.91 | 743.31 | 1457.55 | 918.43 | 846.36 | 432.27 | 1042.23 | 867.43 |

TABLE VI
CLUSTERING PERFORMANCE (%) ON THE BERLIN DATASET. THE BEST RESULTS ARE MARKED IN BOLD

| Class | LMVSC | EOMSC-CA | BMVC | FPMVS-CAG | SMVSC | DCCA | AE2Net | MDC | TMPCC | Ours |
|---|---|---|---|---|---|---|---|---|---|---|
| Forest | 64.96 | 78.22 | 81.88 | 88.72 | 87.90 | 67.55 | 72.44 | 71.45 | **89.47** | 75.17 |
| Residential Area | 43.47 | 39.32 | 32.81 | 52.28 | 50.51 | **70.82** | 46.98 | 46.02 | 40.37 | 43.66 |
| Industrial Area | 9.36 | 16.03 | 49.19 | 42.35 | 4.07 | 4.25 | 25.46 | 28.53 | **63.29** | 62.45 |
| Low Plants | 43.97 | 47.57 | 49.22 | 47.24 | 46.69 | 40.24 | 38.63 | 40.04 | 81.05 | **83.10** |
| Soil | 43.23 | **85.23** | 0.00 | 0.00 | 0.00 | 1.41 | 80.44 | 0.00 | 75.42 | 83.21 |
| Allotment | 0.51 | 2.03 | 62.48 | 2.14 | 4.61 | 0.00 | 1.13 | 6.14 | 0.00 | **65.04** |
| Commercial Area | **52.13** | 39.82 | 6.41 | 1.52 | 33.57 | 47.71 | 28.10 | 22.21 | 36.05 | 28.66 |
| Water | **75.75** | 74.70 | 5.13 | 0.00 | 0.49 | 11.36 | 0.00 | 33.56 | 14.97 | 30.70 |
| ACC | 44.33 | 45.18 | 39.21 | 48.66 | 47.67 | **57.01** | 46.28 | 43.21 | 51.89 | 54.32 |
| Kappa | 26.54 | 31.44 | 25.52 | 29.14 | 24.01 | 33.16 | 27.55 | 24.58 | 40.09 | **42.67** |
| NMI | 29.19 | 31.37 | 28.87 | 22.60 | 22.66 | 34.29 | 29.92 | 28.70 | 39.83 | **43.76** |
| ARI | 18.15 | 18.95 | 12.73 | 14.37 | 13.87 | **30.79** | 20.62 | 18.70 | 27.09 | 28.40 |
| Purity | 73.04 | 73.13 | 67.47 | 57.81 | 57.81 | 70.63 | 73.99 | 67.95 | 76.28 | **79.33** |
| AA | 41.67 | 47.87 | 35.90 | 29.30 | 28.50 | 30.42 | 36.65 | 31.00 | 50.08 | **59.00** |
| Time(s) | 708.02 | 3701.12 | 1542.42 | 18750.25 | 2827.12 | 756.48 | 4977.24 | 4221.32 | 15470.08 | 2538.85 |

overall clustering level. However, the class-level results also reveal an important limitation. Although HCFSN performs well on several dominant or distinctive categories, such as trees, mixed ground surface, road, and building, it obtains 0.00% accuracy on some categories, including mostly grass, dirt and sand, water, building shadow, sidewalk, yellow curb, and cloth panels.

This phenomenon suggests that the learned clusters are not uniformly aligned with all semantic categories. For small categories such as water, yellow curb, and cloth panels, the limited

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

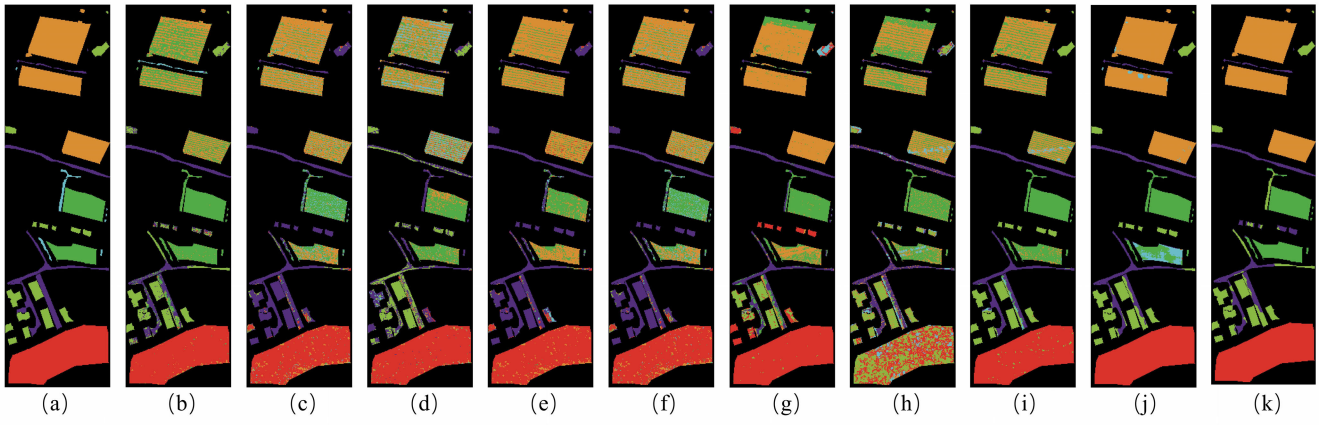



Fig. 5. Clustering maps for the Trento dataset obtained by different clustering approaches. (a) GT. (b) LMVSC. (c) EOMSC. (d) BMVC. (e) FPMVS. (f) SMVSC. (g) DCCA. (h) AE2Nets. (i) MDC. (j) TMPCC. (k) Ours.

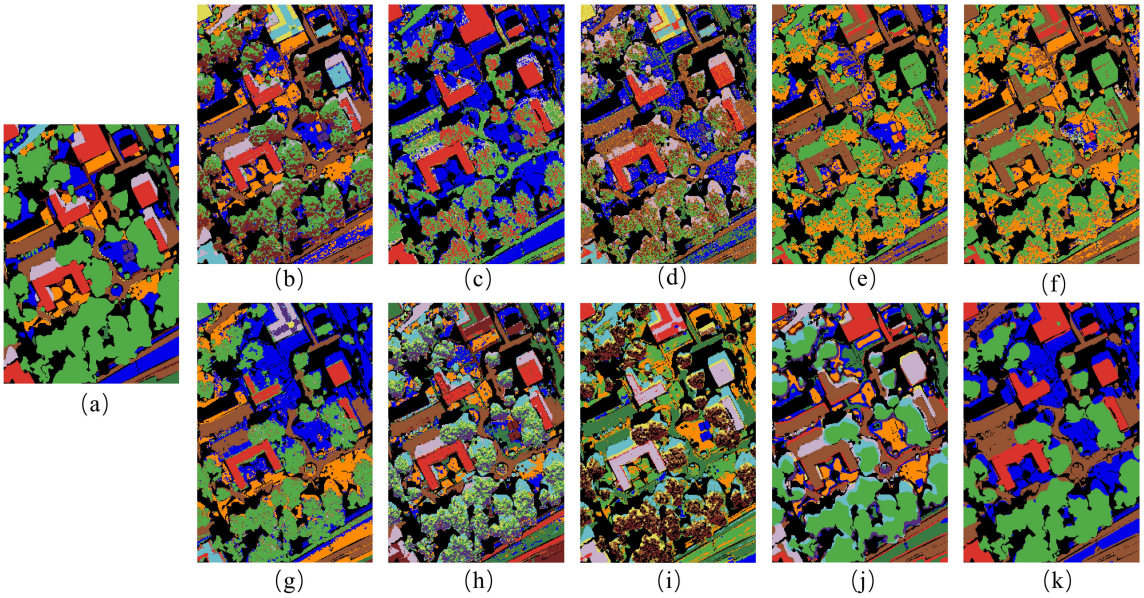



Fig. 6. Clustering maps for the MUUFL dataset obtained by different clustering approaches. (a) GT. (b) LMVSC. (c) EOMSC. (d) BMVC. (e) FPMVS. (f) SMVSC. (g) DCCA. (h) AE2Nets. (i) MDC. (j) TMPCC. (k) Ours.

number of samples makes it difficult for an unsupervised objective to form independent and stable clusters. For homogeneous or visually similar categories such as mostly grass, dirt and sand, building shadow, and sidewalk, the spectral–spatial responses may be easily absorbed by nearby dominant classes after Hungarian matching. Therefore, the 0.00% class accuracy should be interpreted as class-level collapse or category absorption rather than the absence of sample assignments. This explains why HCFSN can still obtain strong overall ACC and ARI while showing limited average class accuracy on MUUFL. The clustering map in Fig. 6 also supports this observation, where several small or ambiguous categories are visually mixed with surrounding regions. These results indicate that HCFSN improves overall multimodal clustering consistency, but still faces challenges in preserving minority and homogeneous class structures. Introducing class-balance constraints, prototype-balance regularization, or cluster-size-aware objectives will be considered in future work to reduce the risk of class collapse.

Table VI compares the clustering performance on the Berlin dataset, with visual results presented in Fig. 7. Different from the Trento and MUUFL datasets, Berlin involves HSI–SAR multimodal fusion, where SAR data provide microwave backscattering information rather than the elevation-related geometric cues offered by LiDAR. This modality difference makes cross-modal semantic correspondence more challenging. HCFSN achieves the best kappa, NMI, purity, and AA on this dataset, indicating that it can still learn useful multimodal representations and preserve relatively balanced class-level clustering performance. However, its OA is lower than that of DCCA, and its ARI is also not the best. This suggests that the proposed hierarchical interaction modules, which are mainly designed to exploit spectral–geometric complementarity in HSI–LiDAR data, may be less optimal for HSI–SAR fusion.

The performance gap on Berlin may be related to several factors. First, SAR data contain speckle noise and backscattering characteristics that are substantially different from LiDAR-derived elevation features, making the LPS and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

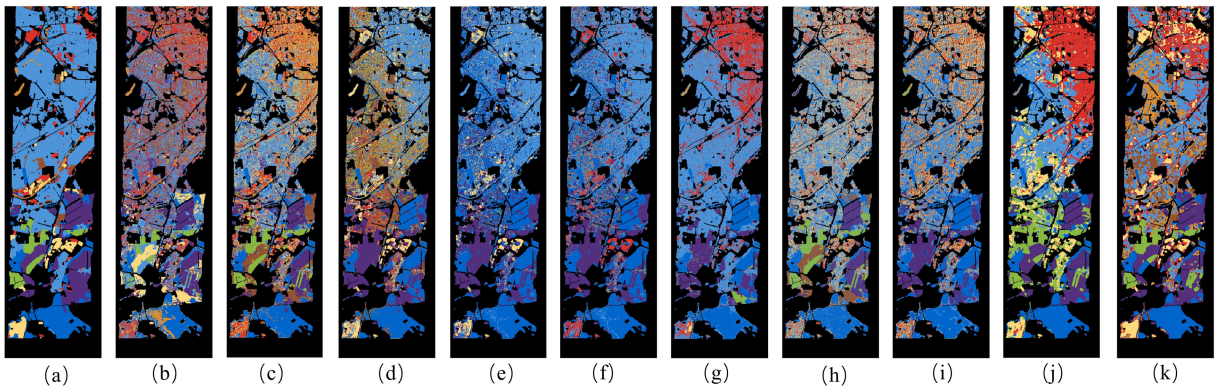



Fig. 7. Clustering maps for the Berlin dataset obtained by different clustering approaches. (a) GT. (b) LMVSC. (c) EOMSC. (d) BMVC. (e) FPMVS. (f) SMVSC. (g) DCCA. (h) AE2Nets. (i) MDC. (j) TMPCC. (k) Ours.

TABLE VII

SYSTEMATIC ABLATION STUDY OF MODULE CONFIGURATIONS ON THE TRENTO AND MUUFL DATASETS

| Variant | Trento | | | | | MUUFL | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | **ACC** | **Kappa** | **NMI** | **ARI** | **Purity** | **ACC** | **Kappa** | **NMI** | **ARI** | **Purity** |
| LPS only | 87.80 | 83.62 | 84.30 | 86.10 | 87.80 | 58.40 | 46.70 | 42.80 | 41.20 | 58.40 |
| DGS only | 89.15 | 85.40 | 85.70 | 88.20 | 89.15 | 60.30 | 49.10 | 44.60 | 43.50 | 60.30 |
| GCMA only | 90.60 | 87.20 | 87.40 | 90.10 | 90.60 | 62.80 | 51.90 | 46.80 | 46.20 | 62.80 |
| w/o LPS | 92.85 | 90.45 | 89.50 | 93.30 | 92.85 | 65.60 | 55.30 | 48.90 | 51.20 | 65.60 |
| w/o DGS | 92.10 | 89.50 | 88.80 | 92.40 | 92.10 | 64.90 | 54.20 | 48.10 | 50.00 | 64.90 |
| w/o GCMA | 91.40 | 88.60 | 87.90 | 91.50 | 91.40 | 63.70 | 52.80 | 47.20 | 48.60 | 63.70 |
| w/o DPA | 93.20 | 90.90 | 89.80 | 93.80 | 93.20 | 66.20 | 56.10 | 49.30 | 52.10 | 66.20 |
| Full model | **94.53** | **92.69** | **91.08** | **95.54** | **94.53** | **68.05** | **58.32** | **50.49** | **54.55** | **68.05** |

channel–group interaction less directly aligned with the physical correspondence between modalities. Second, the Berlin scene has a larger spatial extent and lower spatial resolution, which increases the possibility of mixed land-cover pixels and weakens fine-grained local spatial structures. Third, some urban categories in Berlin, such as residential and commercial areas, may share similar spectral responses and SAR scattering patterns, leading to more ambiguous cluster boundaries. These observations indicate that HCFSN generalizes competitively to HSI–SAR clustering, but its current module design is more suitable for HSI–LiDAR fusion. Designing SAR-aware interaction modules will be considered in future work.

### D. Ablation Study

*1) Effectiveness of Different Modules:* To more clearly evaluate the contribution of each component, we conduct a systematic ablation study on two representative HSI–LiDAR datasets, i.e., Trento and MUUFL. Following the reviewer's suggestion, both single-module variants and module-removal variants are considered. The single-module variants retain only one interaction module after the modality-specific encoders, including LPS only, DGS only, and GCMA only. The module-removal variants remove one component from the full HCFSN while keeping the other modules unchanged, including w/o LPS, w/o DGS, w/o GCMA, and w/o DPA. In particular, w/o DPA removes the DPA refinement after GCMA and directly feeds the bidirectional GCMA output into the clustering head.

As shown in Table VII, the full HCFSN achieves the strongest or highly competitive overall performance on the two representative datasets. Compared with the single-module variants, the full model obtains clearly better clustering results, indicating that LPS, DGS, and GCMA capture complementary information at different interaction levels. Specifically, LPS mainly provides weak local perturbation and improves local robustness; DGS enhances adaptive channel–group semantic exchange; and GCMA further introduces global cross-modal dependency modeling. The removal variants also show that each component contributes to the final performance. Removing LPS weakens local regularization, removing DGS reduces groupwise interaction, and removing GCMA limits long-range cross-modal modeling. In addition, the performance drop of w/o DPA confirms that DPA provides an independent feature-refinement effect by suppressing redundant channel and spatial responses before clustering. These results demonstrate that HCFSN benefits from the hierarchical cooperation of the four modules rather than a simple accumulation of independent components.

*2) Impact of Loss Function:* Table VIII compares three loss configurations on the Trento, MUUFL, and Berlin datasets. The joint loss generally achieves stronger or more stable clustering performance than using either objective alone, indicating that contrastive representation learning and clustering optimization are complementary in the proposed framework.

The loss ablation further clarifies the role of each objective. Using only the contrastive loss improves assignment consistency between augmented views, but it lacks an explicit mechanism to refine cluster prototypes. Using only the clustering loss directly optimizes the assignment structure, but

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
ABLATION STUDY FOR THE LOSS FUNCTIONS OF HCFSN ON THE TRENTO, MUUFL, AND BERLIN DATASETS

| | Trento | | | | | MUUFL | | | | | Berlin | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **ACC** | **Kappa** | **NMI** | **ARI** | **Purity** | **ACC** | **Kappa** | **NMI** | **ARI** | **Purity** | **ACC** | **Kappa** | **NMI** | **ARI** | **Purity** |
| Con loss only | 84.08 | 79.18 | 88.17 | 85.01 | 89.51 | 52.03 | 43.27 | **53.14** | 36.44 | **71.50** | 54.31 | 37.60 | 38.39 | 37.35 | 70.22 |
| Clu loss only | 76.34 | 76.47 | 79.43 | 78.30 | 76.34 | 43.30 | 41.16 | 41.35 | 45.68 | 43.30 | 46.53 | 32.88 | 39.25 | **44.34** | 52.36 |
| Joint loss | **94.53** | **92.69** | **91.08** | **95.54** | **94.53** | **68.05** | **58.32** | 50.49 | **54.55** | 68.05 | **54.32** | **42.67** | **43.76** | 28.40 | **79.33** |

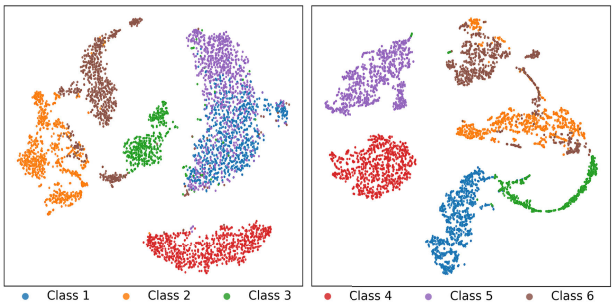



Fig. 8. Feature-space visualization on the Trento dataset using t-SNE. Different colors denote different reference categories, which are used only for visualization after training. (a) Raw multimodal features. (b) Learned embeddings produced by HCFSN.

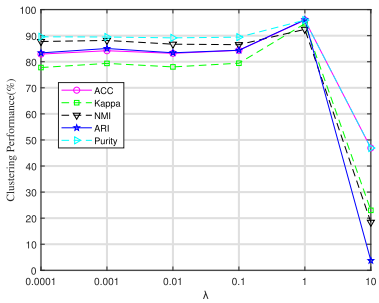



Fig. 9. Sensitivity analysis of parameter $\lambda$ on the Trento dataset.

the learned assignments are less robust to augmentation-induced variations. The joint objective achieves better overall performance because the assignment-level contrastive loss and the prototype-based clustering loss optimize the clustering process from complementary perspectives.

### E. Feature Separability Analysis

To further verify whether HCFSN learns discriminative multimodal representations, we conduct feature-space visualization and quantitative separability analysis on the representative Trento dataset. Before computing the separability measures, both raw features and learned embeddings are normalized to avoid scale effects.

Fig. 8 shows the t-SNE visualization of raw multimodal features and the learned embeddings produced by HCFSN. Compared with the raw features, the learned embeddings exhibit more compact intraclass distributions and clearer interclass separation. This indicates that HCFSN improves the discriminability of multimodal representations rather than merely changing the final clustering assignments.

TABLE IX
FEATURE SEPARABILITY ANALYSIS ON THE TRENTO DATASET

| Feature type | $C_{\text{intra}} \downarrow$ | $C_{\text{inter}} \uparrow$ | Ratio↑ |
|---|---|---|---|
| Raw multimodal features | 0.3415 | 1.5528 | 4.5476 |
| HCFSN embeddings | **0.0262** | **1.8214** | **69.6460** |

TABLE X
COMPUTATIONAL COMPLEXITY ANALYSIS OF HCFSN ON THE TRENTO DATASET

| Method | Params (M) | FLOPs (G) | Train (s) | Infer. (ms/sample) | Mem. (GB) |
|---|---|---|---|---|---|
| HCFSN | 1.49 | 0.18 | 1028.21 | 0.19 | 2.10 |

As shown in Table IX, HCFSN achieves a much lower intracluster compactness than the raw multimodal features, decreasing $C_{\text{intra}}$ from 0.3415 to 0.0262. Meanwhile, the intercluster separation increases from 1.5528 to 1.8214. As a result, the separation ratio is substantially improved from 4.5476 to 69.6460. These results further demonstrate that the learned embeddings are more compact and better separated, confirming the representation-level effectiveness of the proposed framework.

### F. Computational Complexity Analysis

To further analyze the computational cost of HCFSN, we report a computational complexity analysis on the representative Trento dataset. Since several compared baselines are traditional nonneural subspace clustering methods, their network parameters and FLOPs are not directly comparable with those of deep neural models. Therefore, we retain the runtime comparison of all methods in the main quantitative tables and provide a dedicated complexity analysis for HCFSN. All measurements are conducted under the same RTX 3090 experimental setting. The number of parameters is calculated from the trainable model variables, FLOPs are counted for one $7 \times 7$ multimodal input sample, and GPU memory denotes the peak training memory.

As shown in Table X, HCFSN contains 1.49M trainable parameters and requires 0.18G FLOPs for one multimodal input sample. The full training time on the Trento dataset is 1028.21 s, and the inference time is 0.19 ms per sample. The peak GPU memory consumption is 2.10 GB. These results show that although HCFSN introduces hierarchical cross-modal interaction and global attention, its model size remains moderate and the computational cost is manageable for patch-based multimodal clustering. The additional cost mainly comes from the global cross-modal attention and attention-based

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

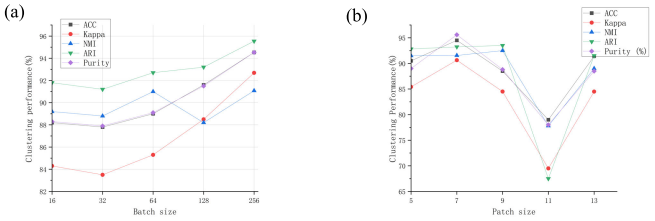



Fig. 10. (a) Influence of batch size. (b) Influence of patch size.

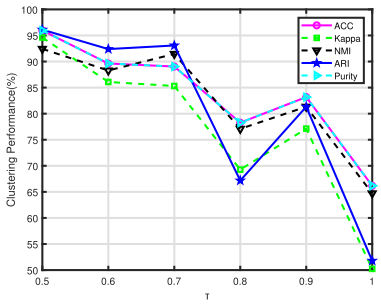



Fig. 11. Behavior of temperature coefficient on the Trento dataset.

feature refinement modules. Considering the improved clustering performance and feature separability, HCFSN provides an acceptable tradeoff between representation quality and computational efficiency.

### G. Parameter Sensitivity

To examine the influence of key hyperparameters, we conduct parameter sensitivity analysis on the representative Trento dataset. This analysis is used to evaluate the robustness of HCFSN to different training configurations and to provide a reasonable basis for the final parameter settings. After the hyperparameters are determined, the same settings are used in the main experiments on all datasets, rather than being tuned separately for each dataset.

*1) Sensitivity of $\lambda$:* Fig. 9 investigates how the balance parameter $\lambda$ affects clustering performance on the Trento dataset, with $\lambda$ ranging from 0.0001 to 10 on a logarithmic scale. The results show that $\lambda = 1$ provides a good balance between assignment-level contrastive consistency and clustering optimization. When $\lambda$ is too small, the clustering objective is insufficiently emphasized. When $\lambda$ is too large, the optimization becomes dominated by the clustering term, which may weaken the consistency of the learned assignment distributions. Therefore, we set $\lambda = 1$ in the main experiments and keep it unchanged across datasets.

*2) Impact of the Size of Batch and Patch:* Fig. 10 presents a systematic analysis of two critical hyperparameters affecting HCFSN's clustering performance: batch size and patch size. The results show that both batch size and patch size affect the stability of HCFSN. A moderate-to-large batch size generally provides more stable contrastive optimization, while an excessively small batch may provide insufficient negative samples for assignment-level contrastive learning. For patch size, $7 \times 7$ achieves a favorable balance between local spatial context and mixed-class interference. Smaller patches may not contain enough spatial context, whereas larger patches may introduce pixels from different land-cover categories and weaken cluster purity. Therefore, we use a batch size of 256 and a patch size of $7 \times 7$ in the main experiments.

*3) Behavior of Temperature Coefficient:* Fig. 11 examines the effect of the temperature coefficient $\tau$ on the clustering performance of HCFSN. The results show that $\tau$ affects the sharpness of the assignment-level contrastive objective. A smaller $\tau$ produces sharper assignment discrimination, but an excessively small value may make the optimization sensitive to hard negative samples. A larger $\tau$ smooths the assignment distribution, but may reduce the discriminability of cluster assignments. Considering both assignment discriminability and training stability, we set $\tau = 0.5$ in all main experiments.

## V. Conclusion

In this article, we proposed a HCFSN for unsupervised multimodal RS clustering. HCFSN integrates LPS, DGS, GCMA, and DPA to model cross-modal interactions from local, channel–group, global, and refinement levels. The experimental results on Trento, MUUFL, and Berlin show that HCFSN achieves competitive and generally strong performance on most major clustering metrics, suggesting the effectiveness of hierarchical cross-modal interaction for multimodal representation learning.

Despite these results, this work still has several limitations. First, the current framework is mainly designed for HSI–LiDAR fusion, and its generalization to HSI–SAR data remains less stable, as reflected by the nonbest OA on the Berlin dataset. Second, the class-level results on MUUFL show that minority or homogeneous categories may be absorbed by dominant clusters, indicating that class-balance preservation remains challenging in unsupervised clustering. Third, the computational complexity of multiscale fusion increases with input size, which may restrict scalability to extremely large scenes. Future work will explore SAR-aware interaction modules, class-balance or prototype-balance constraints, and lightweight attention mechanisms for more efficient multimodal clustering.

## References

[1] A. Ma et al., "Fast sequential feature extraction for recurrent neural network-based hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5920–5937, Jul. 2021, doi: 10.1109/TGRS.2020.3018449.

[2] M. Khodadadzadeh, J. Li, S. Prasad, and A. Plaza, "Fusion of hyperspectral and LiDAR remote sensing data using multiple feature learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2971–2983, Jun. 2015.

[3] B. Rasti, P. Ghamisi, J. Plaza, and A. Plaza, "Fusion of hyperspectral and LiDAR data using sparse and low-rank component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6354–6365, Nov. 2017.

[4] X. Zheng, H. Sun, X. Lu, and W. Xie, "Rotation-invariant attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 4251–4265, 2022.

[5] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[6] L. Gómez-Chova, D. Tuia, G. Moser, and G. Camps-Valls, "Multimodal classification of remote sensing images: A review and future directions," *Proc. IEEE*, vol. 103, no. 9, pp. 1560–1584, Sep. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.

[7] C. Ge, Q. Du, W. Li, Y. Li, and W. Sun, "Hyperspectral and LiDAR data classification using kernel collaborative representation based residual fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1963–1973, Jun. 2019.

[8] W. Li, J. Wang, Y. Gao, M. Zhang, R. Tao, and B. Zhang, "Graph-feature-enhanced selective assignment network for hyperspectral and multispectral data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526914.

[9] Y. Gao, M. Zhang, W. Li, and R. Tao, "Distribution-independent domain generalization for multisource remote sensing classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 13333–13344, Jul. 2025.

[10] M. Pedergnana, P. R. Marpu, M. Dalla Mura, J. A. Benediktsson, and L. Bruzzone, "Classification of remote sensing optical and LiDAR data using extended attribute profiles," *IEEE J. Sel. Topics Signal Process.*, vol. 6, no. 7, pp. 856–865, Nov. 2012.

[11] R. Huang and J. Zhu, "Using random forest to integrate LiDAR data and hyperspectral imagery for land cover classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, 2013, pp. 3978–3981.

[12] C. Demirkesen, M. Teke, and U. Sakarya, "Hyperspectral images and LiDAR based DEM fusion: A multi-modal landuse classification strategy," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 2942–2945.

[13] J. Xia, N. Yokoya, and A. Iwasaki, "A novel ensemble classifier of hyperspectral and LiDAR data using morphological features," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2017, pp. 6185–6189.

[14] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[15] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[16] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[17] X. Zhao, R. Tao, W. Li, W. Philips, and W. Liao, "Fractional Gabor convolutional network for multisource remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[18] Y. Gao et al., "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[19] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[20] L. Zhou, J. Geng, and W. Jiang, "Joint classification of hyperspectral and LiDAR data based on position-channel cooperative attention network," *Remote Sens.*, vol. 14, no. 14, p. 3247, Jul. 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/14/3247

[21] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716, doi: 10.1109/TGRS.2022.3232498.

[22] M. Wang, Y. Sun, J. Xiang, R. Sun, and Y. Zhong, "Joint classification of hyperspectral and LiDAR data based on adaptive gating mechanism and learnable transformer," *Remote Sens.*, vol. 16, no. 6, p. 1080, Mar. 2024. [Online]. Available: https://www.mdpi.com/2072-4292/16/6/1080

[23] H. Pan, X. Li, H. Ge, L. Wang, and C. Shi, "A hierarchical coarse-fine adaptive fusion network for the joint classification of hyperspectral and LiDAR data," *Remote Sens.*, vol. 16, no. 21, p. 4029, 2024. [Online]. Available: https://www.mdpi.com/2072-4292/16/21/4029

[24] T. Chen, S. Kornblith, M. Norouzi, and G. E. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. 37th Int. Conf. Mach. Learn.*, vol. 119, 2020, pp. 1597–1607.

[25] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.

[26] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[27] E. D. Cubuk, B. Zoph, D. Mane, V. Vasudevan, and Q. V. Le, "AutoAugment: Learning augmentation policies from data," 2018, *arXiv:1805.09501*.

[28] J.-B. Grill et al., "Bootstrap your own latent-a new approach to self-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21271–21284.

[29] X. Chen and K. He, "Exploring simple Siamese representation learning," in *Proc. IEEE Comput. Soc. Conf. Comput. Vision Pattern Recognit.*, Jun. 2021, pp. 15750–15758.

[30] J. Žbontar, J. Li, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 12310–12320.

[31] Y. Li, P. Hu, Z. Liu, D. Peng, J. T. Zhou, and X. Peng, "Contrastive clustering," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 10, pp. 8547–8555.

[32] Y. Cai, Z. Zhang, P. Ghamisi, B. Rasti, X. Liu, and Z. Cai, "Transformer-based contrastive prototypical clustering for multimodal remote sensing data," *Inf. Sci.*, vol. 649, Nov. 2023, Art. no. 119655.

[33] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[34] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 3011–3024, Jun. 2017.

[35] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6000–6010.

[36] X. Sun et al., "From single- to multi-modal remote sensing imagery interpretation: A survey and taxonomy," *Sci. China Inf. Sci.*, vol. 66, no. 4, Apr. 2023, Art. no. 140301.

[37] C. Gao et al., "MIFTr: Efficient universal image matching with modality-invariant feature transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5510817, doi: 10.1109/TGRS.2026.3683096.

[38] Y. Ding et al., "Self-supervised locality preserving low-pass graph convolutional embedding for large-scale hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536016.

[39] Y. Tian, D. Krishnan, and P. Isola, "Contrastive multiview coding," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 776–794.

[40] Y. Cai, X. Liu, and Z. Cai, "BS-nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, Mar. 2020.

[41] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[42] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[43] S. Wang et al., "Fast parameter-free multi-view subspace clustering with consensus anchor guidance," *IEEE Trans. Image Process.*, vol. 31, pp. 556–568, 2022.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:05:51 UTC from IEEE Xplore. Restrictions apply.